

Nikita Kazeev

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Research Scientist at the intersection of AI and Physics

WORK EXPERIENCE

PERFOMAX <i>AI Advisor</i>	2026–present
CONSTRUCTOR GROUP <i>Machine Learning Consultant</i>	2021–present
NATIONAL UNIVERSITY OF SINGAPORE <i>Postdoc under Kostya Novoselov</i>	2022–present

Leadership

- Program Chair for the [AI4X 2025/6 conferences](#)
- Main organiser of the [ICLR 2025 Workshop on Machine Learning Multiscale Processes](#)
- Co-PI of a US\$3.4m AI Singapore grant on multiscale machine learning

Wyckoff Transformer, a generative model for materials [ICML 2025](#) · [GitHub](#)

- Developed a coordinate-free permutation-invariant autoregressive Transformer encoder model for material generation and property prediction based on symmetry inductive bias
- Increased materials discovery yield (S.U.N.) by 1.24x
- Cut inference to 0.05 GPU-ms/structure — 4 orders of magnitude faster than prior models
- Implemented the model in PyTorch; production-ready package published on [PyPI](#)
- Orchestrated large-scale (10k structures) DFT computations with VASP on an HPC cluster
- Led a team of 6

ML for defects in 2D crystals [npj 2D Mater. Appl.](#) · [npj Comput. Mater.](#) · [patent](#) · [GitHub](#)

- Conceived a novel sparse representation of crystals with defects
- Cut energy-prediction error 3.7x vs. the best prior model
- Optimised training and inference: 4x less memory, 8x fewer GPU operations
- Built a reproducible pipeline for parallel ML experiments
- Led a team of 3

(CERN | YANDEX → HSE UNIVERSITY) *Researcher* 2014–2022
[2025 Breakthrough Prize in Fundamental Physics](#) as a member of LHCb.

Generative models uncertainty estimation [paper](#) · [conference](#) · [code 1](#) · [code 2](#)

- Developed the first methods for estimating uncertainty of conditional GANs
- Led a team of 3 students

Generative models for fast simulation [paper 1](#) · [paper 2](#) · [paper 3](#) · [code](#)

- Built a GAN for high-fidelity, tabular-data simulation of a Cherenkov detector
- Reached 10⁵x speed-up over the full Geant4 simulation

Machine learning on noisy data [paper 1](#) · [paper 2](#) · [code](#)

- Derived a statistically rigorous method to train classifiers on sPlot-weighted (label-noisy) data, standard in high-energy physics

Muon identification at the LHCb experiment at CERN [paper](#) · [poster](#) · [IDAO-2019](#)

- Solved a classification problem over tabular data with noisy labels under timing constraints
- Built a CatBoost model that cut the false-positive rate 30% in the critical low-momentum region
- Shipped the model into the LHCb trigger (C++ & Python)
- Packaged the work as a data science competition problem for IDAO-2019

CatBoost aka fighting biases with dynamic boosting [paper](#)

- Built the team's distributed experiment infrastructure (Bash/Python)
- Studied gradient boosting improvements with experiments on toy data

EDUCATION

HSE University & Sapienza Università di Roma <i>PhD in Computer Science and Physics (Double Degree)</i> Supervisors: Andrey Ustyuzhanin & Barbara Sciascia Thesis: Machine Learning for particle identification in the LHCb detector	2016–2020
Product Management course at Yandex <i>Focused coursework in product strategy and execution.</i>	2018
Yandex School of Data Analysis <i>Master's-level CS programme</i> Algorithms, machine learning, deep learning, and distributed systems.	2013–2015
Moscow Institute of Physics and Technology <i>MS in Physics</i> Optimisation of data processing of the LHCb experiment.	2014–2016
<i>BS in Physics</i> Wave-packet molecular dynamics of electrons in nonideal plasma.	2010–2014
International Junior Science Olympiad (IJSO) Silver medal	Korea, 2008

MENTORSHIP

- Mentored 9 students ([1](#), [2](#), [3](#), [4](#), [5](#), [6](#), [7](#), [8](#), [9](#)), 3 interns, and student workshop projects ([1](#), [2](#)).

TEACHING & OUTREACH

- Pitch research to the general public through major newspaper coverage ([1](#), [2](#)), blog posts ([1](#), [2](#)), a [science slam](#), interviews ([1](#), [2](#)), and a public [talk](#).
- Taught at the Machine Learning in High Energy Physics Summer School in [2015](#), [2016](#), [2017](#), [2018](#), [2019](#), [2020](#), [2021](#), and [2023](#).
- Lecturer in the [award-winning](#) Coursera Advanced Machine Learning [specialisation](#)
- Taught and developed an introductory data analysis [course](#) at HSE.
- Delivered more than 20 [conference talks](#).

SERVICE

- Reviewer for NeurIPS, RSC Advances, Machine Learning: Science and Technology, Digital Discovery
- Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

SKILLS

- **Generative modelling for scientific data** — GANs, autoregressive transformers, and diffusion; from fast detector simulation to symmetry-aware crystal generation.
- **Uncertainty quantification** — built the first methods for estimating the uncertainty of conditional GANs; calibrated generative and discriminative models.
- **ML on structured, non-text data** — tabular, graph, and atomistic representations, spanning gradient boosting to transformers.
- **Inference and systems at scale** — C++ production integration, large-scale DFT/VASP on HPC clusters, and reproducible experiment pipelines.
- **Physics domain depth** — high-energy and condensed-matter physics; first-principles intuition that shapes model design.
- **Research leadership** — lead multidisciplinary teams, chair conferences, and mentor students from idea to publication.

ML & AI • Python • PyTorch • C++ • HPC • Physics • Research Writing & Speaking • Leadership & Mentorship

PRESENTING AT ICML 2026

July 11, 12:15–13:30 — Nikita Kazeev and Ian Babich. “[Foresight-Phys: A Benchmark for Forecasting the Results of Physical Experiments](#)”. In: *Forecasting as a New Frontier of Intelligence Workshop at ICML 2026*.

July 11, 15:00–16:30 — Nikita Kazeev and Phan Bui Nhat Huyen. “[Position: Align AI to Our Aspirations, Not Our Flaws](#)”. In: *Pluralistic Alignment Workshop at ICML 2026*.

July 11, 15:30–17:00 — Nikita Kazeev and Andrey Ustyuzhanin. “[The Geometry of Reasoning Failure: Predicting Agent Errors from Semantic Scale Trajectories](#)”. In: *Statistical Frameworks for Uncertainty in Agentic Systems Workshop at ICML 2026*.

July 11, 10:50–11:50 & 15:55–16:55 — Artem Dembitskiy*, Shuya Yamazaki*, Artem Maevskiy*, **Nikita Kazeev***, Roman A. Eremin, Semen Budennyi, Kedar Hippalgaonkar, Antonio Helio Castro Neto, and Andrey E. Ustyuzhanin. “[Generative Pipeline for Discovering Solid-State Battery Materials with Universal Atomistic Potentials](#)”. In: *AI for Science Workshop at ICML 2026*. *Equal Contribution.

July 10, 11:40–13:30 & 16:10–17:00 — Nikita Kazeev and Andrey Ustyuzhanin. “[Uncertainty Quantification for LLM Agents via Semantic Abstraction Trajectories](#)”. In: *Structured Probabilistic Inference & Generative Modeling Workshop at ICML 2026*.

SELECTED PUBLICATIONS

Nikita Kazeev, Wei Nong, Ignat Romanov, Ruiming Zhu, Andrey E Ustyuzhanin, Shuya Yamazaki, and Kedar Hippalgaonkar. “[Wyckoff Transformer: Generation of Symmetric Crystals](#)”. In: *Proceedings of the 42nd International Conference on Machine Learning*. Ed. by Aarti Singh, Maryam Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff, and Jerry Zhu. Vol. 267. Proceedings of Machine Learning Research. PMLR, 13–19 Jul 2025, pp. 29495–29526.

Nikita Kazeev, A. R. Al-Maeni, I. Romanov, et al. “[Sparse representation for machine learning the properties of defects in 2D materials](#)”. In: *npj Computational Materials* 9 (2023), p. 113.

Maxim Borisyak and **Nikita Kazeev**. “[Machine Learning on data with sPlot background subtraction](#)”. In: *Journal of Instrumentation* 14.08 (2019). Alphabetic order, I’m the corresponding author.

Denis Derkach, **Nikita Kazeev**, Fedor Ratnikov, Andrey Ustyuzhanin, and Alexandra Volokhova. “[Cherenkov detectors fast simulation using neural networks](#)”. In: *Nuclear Instruments and Methods in Physics Research Section A* (2019). Alphabetic order, I’m the corresponding author.